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Comprehensive assessment of the water environment carrying capacity based on the spatial system dynamics model, a case study of Yongding River Basin in North China

Dan Dai [a,b], Mingdong Sun [a], Xubo Lv [a], Juxiang Hu [b], Hua Zhang [a], Xiangqin Xu [a], Kun Lei [a,*]

[a] State Key Laboratory of Environmental Criteria and Risk Assessment, Chinese Research Academy of Environmental Sciences, Beijing, 100012, PR China
[b] Key Laboratory of Ecological Impacts of Hydraulic-Projects and Restoration of Aquatic Ecosystem of Ministry of Water Resources, Institute of Hydroecology, Ministry of Water Resources and Chinese Academy of Sciences, Wuhan, 430079, PR China

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ABSTRACT

The Chinese water environment system is severely degraded by intensive anthropogenic activities. To relieve these problems, the national government has proposed to establish a comprehensive evaluation system of water environment carrying capacity (WECC). However, few studies provide a comprehensive depiction of the spatiotemporal heterogeneity of WECC within a basin/region owing to the limitations of interdisciplinary research methods. By integrating the system dynamics model, cellular automaton model, and gridded geographic information system technology, we developed a spatial system dynamics model to explore the spatiotemporal pattern of the WECC of Yongding River Basin in North China. The spatial system dynamics model works well and the simulation accuracy was close to 85%. Our results showed that inadequate water resources and water quality induced the overloaded WECC in Yongding River Basin, which was unevenly distributed within the river basin. From 2017 to 2035, Zhangjiakou often suffered an overload of WECC throughout the year due to quantity-related water shortages. However, in Beijing's mountainous area, the WECC status of control units 1 and 4 was in good status owing to sufficient water quantities. Considering the overload of the WECC and its feedback relationship with anthropogenic activities, effective and feasible management measures, such as reducing agricultural irrigation water use and domestic pollution, improving the rate of sewage collection and treatment, or industrial restructuring should be prioritized in this arid river basin. Our study improved methodologies in WECC modeling for river basins, helped overcome difficulties in revealing the spatiotemporal dynamic patterns of WECC, proposed a calculation method of WECC in terms of population and GDP, and verified that the threshold of WECC was not static, which provides information for finding a balance between economic development and water safety, especially in regions with water scarcity.

1. Introduction

Water is the lifeblood of the biosphere, an important supporter of the sustainable development of nature and society, one of the most precious natural resources on earth, yet also susceptible to the adverse effects of human activities (He et al., 2021; Vörösmarty et al., 2010). Ensuring water safety is one of the 17 Sustainable Development Goals established by the United Nations (Griggs et al., 2013). In recent decades, China's rapid development has caused serious water environment problems, which have attracted the focus of the international research community (Barnett et al., 2015; Zhao et al., 2015), although the national inland surface water-related problems having been improved (Gu et al., 2017; Ma et al., 2020).

In light of these problems, the water environment carrying capacity (WECC) has been developed to measure sustainable water development (Zhao et al., 2021; Zhou et al., 2017, 2019) based on the theory of carrying capacity. The concept of the WECC can be defined as "the largest population and economic scale that the water environment system can support in a specific basin or region during a period while meeting its own health needs (Zhou et al., 2017)." According to this, two major systems are involved in the study of the WECC, the human system and the water environment system. Research on the WECC should not only be concerned with the resources and environmental attributes of the water environment system but more with the dynamic coordination

* Corresponding author.
E-mail address: leikun@craes.org.cn (K. Lei).

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between the human system and the water environment system. The essence of the WECC is to seek the balance between the water environment system and socio-economic development. Additionally, considering that land use affects the spatial pattern of the WECC, as it is the bond of interaction and connection between human activities and water environment processes, research of WECC can be seen as a complicated function related to the time and space of population, economy, land use change, technology, policy, water resources, water environment, etc. (Zhou et al., 2017). However, the complicated interaction between human activities and natural water processes, their feedback mechanisms, and the diversity of influencing factors lend an inherent complexity to WECC modeling.

Currently, WECC research has not yet formed a unified evaluation method and is still at the theoretical stage. The existing WECC quantification and assessment methods can be classified into two categories based on the assessment timeliness. The first category is the comprehensive evaluation method of the index system and the water footprint (e.g., Ren et al. (2016), Wang and Xu (2015), Hubacek et al. (2009) and Wei et al. (2019)). Such methods are simple but fail to predict the future WECC. The second category is the system dynamics (SD) model and integrated modeling methods. For example, Yang et al. (2015) used the SD model to simulate and predict the WECC of Tieling City, in northeast China. Jin et al. (2009) integrated the SD model into the ecological footprint model to explore urban green development. Wang et al. (2018) integrated the SD model and the analytic hierarchy process to assess the WECC in a lake basin in northwest China. Although the SD model has powerful time dynamic simulation capabilities, it is unable to reveal the spatial characteristics of the subsystems (Ahmad and Simonovic, 2001). Therefore, these combined models largely fail to consider the internal and spatial heterogeneity of the WECC, because the study areas they are applied to are commonly considered as a single homogeneous entity without internal heterogeneity.

The inability of the WECC to reveal the internal spatial heterogeneity of the human and water environment systems means that the research results do not correspond to the reality, which can cause decision-makers to become biased. For example, the WECC may overload in a partial space, but not on the entire scale; alternatively, the WECC may overload on the entire scale, but not in a partial space, which will lead to a disconnection between the results of WECC research and the actual water environment management process to a certain extent. Considering that human and natural systems have a high degree of dynamic complexity and that the elements of these systems have non-linear characteristics and spatial differences (Liu et al., 2007), the continuous changes of the intensity of human activities make the WECC exhibit strong spatiotemporal dynamics. Therefore, analyzing spatiotemporal changes of the WECC in efforts to enrich research on its spatial heterogeneity and dynamic evolution is urgent. To enable the revealing and prediction of the spatiotemporal characteristics of the WECC, we propose a newly developed WECC simulation model based on the SD model, cellular automata (CA) model, and gridded geographic information system (GIS) technology, which considers the spatial heterogeneity of the water environment, as well as socio-economic and land use change conditions.

The aim of this study was to develop a spatial system dynamics model with the following objectives: (1) analyze and assess the WECC under the influence of land use change, (2) reveal the spatiotemporal changes of the human and water environment systems, and (3) quantify the WECC based on the population and GDP instead of using dimensionless values and finally give some potent water management strategies. Our results will fill the knowledge gap in terms of the spatiotemporal heterogeneity of the WECC and predicting future water safety status, thereby providing decision-makers with important information in water scarcity regions to step back from its water crisis. The WECC simulation model developed herein may be useful in water-related sustainability studies.

2. Materials and methods

2.1. Study area and data collection

2.1.1. Study area

The Yongding River is located in the Hai River Basin in North China, and it is the "Mother Water of Beijing City (capital of China)" with a basin area of 47,000 km^2 and a total length of 747 km. It originates from the Yang River in Inner Mongolia and the Sanggan River in Shanxi Province. Our study area extends from Cetian Reservoir in Shanxi Province to Sanjiadian in Beijing Municipality and it is the water conservation area of the northwest Yongding River, with an area of 19612 km^2, including Zhangjiakou City in Hebei Province (17355 km^2) and Beijing (2257 km^2), accounting for 70.77% of the whole river basin in Beijing (Fig. 1). The Yongding River has been dried for a long time in the plain area of Beijing, and the channels served as gravel mines and dumping grounds for sewage and garbage without external water supply (Jiang et al., 2014). At present this region has become a hot spot area across the whole basin because it has an important impact on the water ecological security of the capital city Beijing, and the main water problems are water shortage and water contamination (Dai et al., 2020b; Jiang et al., 2014; Xia et al., 2012). As one of the biggest reservoirs in YRB, the Guanting Reservoir cannot be used as a drinking water source since 1997 owing to extensive pollution (He et al., 2011). This region has long been recognized as one of the most water-deficient areas in China owing to its low water resources (243 m^3 per capita·a^{-1}), which is far below the threshold of absolute water scarcity (500 m^3 per capita·a^{-1}) (Falkenmark et al., 1989). The continued economic growth may undoubtedly further aggravate the pressure of the regional water problems (Fang et al., 2019; Yang et al., 2019). Therefore, it is highly significant to explore the WECC and to alleviate water-related problems in this water-scarce region.

The control unit is a relatively new concept in China's water environment management (Huang et al., 2019; Zhang et al., 2021), which has been put forward to achieve China's basin water quality target management objectives. According to this method, a specific basin or region is divided into several spatial units or sub-basins. Each spatial unit or sub-basin can be seen as a control unit. The division of the control unit takes into consideration the characteristics of the river basins and administrative divisions. Lei et al. (2013) explained the method and specific process of dividing the control unit. We divided this study area into 19 control units based on natural and socio-economic characteristics. Control units 1, 4, and 13 belong to the mountainous area of Beijing, control unit 9 is the Guanting Reservoir,[1] and control units 2, 3, 5 to 8, and 10 to 19 belong to Zhangjiakou City (Fig. 1).

2.1.2. Data collection

The VENSIM software (Vensim DSS) was selected to construct the SD model. Data used in the SD model were mainly collected from the Beijing Municipal and Zhangjiakou socio-economic and environmental quality statistical yearbooks, as well as from the National Hydrological Yearbook. The CA-Markov model in the IDRISI Selva 17 (Clark Lab) platform was selected to construct the CA model. The basic input data in the CA-Markov model included the Digital Elevation Model (DEM) and the map of the ecological red lines, urban extent maps, land use maps, and transportation data of 2000, 2005, and 2015.

2.2. Spatial system dynamics model constructed by coupling the SD, CA, and GIS

The spatial system dynamics (SSD) method was first proposed by Ahmad and Simonovic (2001) and is mainly used to describe the grid-based interaction of the dynamics of modules of spatially

[1] Guanting Reservoir is not included in the scope of this study.

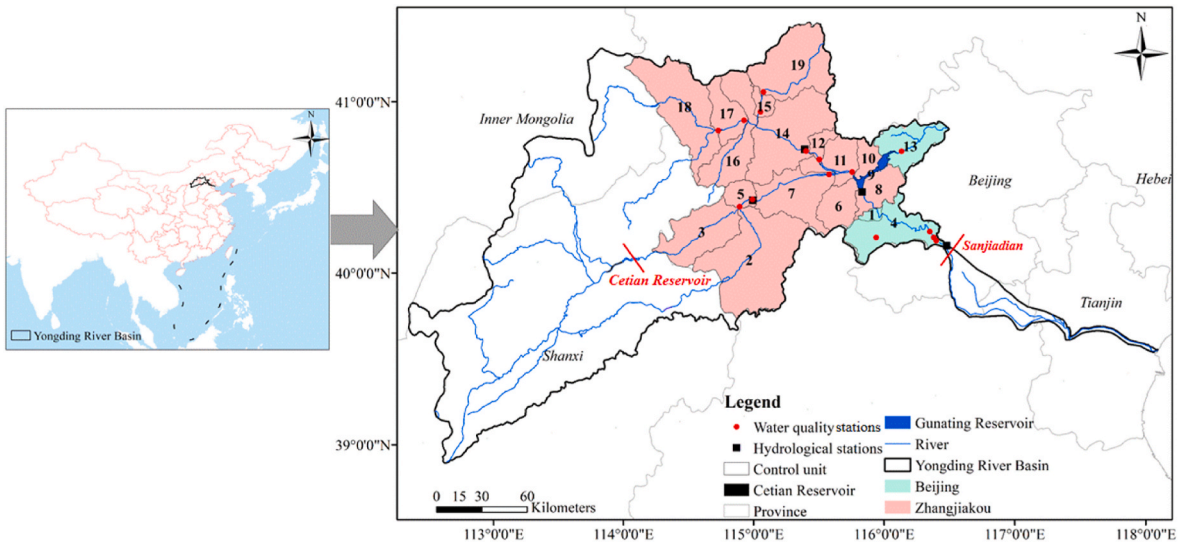


Fig. 1. Location of the control unit and the study area.

distributed systems (Roach and Tidwell, 2009). The SSD model can simulate the changes of non-spatial and geospatial elements over time, and provide feedback based on the complex dynamic process of time and space. In general, two modeling methods were used to construct the SSD model, namely, the SD model and GIS spatial analysis technology (Neuwirth et al., 2015a). The SSD methodology is widely used in environmental simulations. Zhang et al. (2011) constructed the GIS-SD framework to simulate the temporal and spatial changes of water pollutant concentrations. Nikolic et al. (2013) integrated the GIS, SD model, agent-based model, and hydrologic model to simulate the groundwater inflow. Neuwirth et al. (2015b) coupled the SD model to GIS via a Python program to simulate the land cover types. More current studies on water environment systems using SSD models are focused on hydrology and water resources, and relatively few studies have been focused on the WECC.

The SSD model of WECC is established by coupling the interactions of the SD model, the CA model, and the gridded GIS technology (Fig. 2). First, the CA model provides the simulated land use data to drive the SD model. Then the output of the SD model, including population and GDP to create the suitability map in the CA-Markov model to drive the urban land evolution for the two sub-regions, and the simulated data of future land use are used to drive the simulation of the SD model in the next stage. Finally, with the use of gridded GIS technology, the related coefficients of each grid are generated by the predicted land use data exported from the CA-Markov model and the socio-economic data exported from the SD model. By calculating the coefficient of each grid and the amount of water use and pollutant discharge exported from the SD model, the spatial patterns of water use and water pollutants are obtained (Fig. 2).

2.2.1. Construction of the SD model

Owing to differences in the geographical and socio-economic characteristics of the Zhangjiakou and Beijing mountainous areas, the study area was set as $S = \{S_1, S_2\}$, where S_1 and S_2 are the WECC subsystems in Zhangjiakou and Beijing, respectively. Each subsystem consists of five components: population, economy, water resources, land-use change, and pollutant discharge. The simulation parts also involve the water pollutant carrying rate (WPCR), water resources carrying rate (WRCR), and the WECC calculations. The model started in 2006 and ended in 2035, with a one-year simulation time step. Detailed descriptions of the population, economy and water resource modules can be found in our previous paper (Dai et al., 2020a) and the supplementary materials (equations S1-S5). The pollutant discharge components and land-use change are shown below.

(1) Pollutant discharge estimation module

This module aimed to simulate the amount of inflow of pollutants (Fig. S1). It was calculated using the following equation:

$$PD = ULP + RLP + AP + IP + TIP + UNP + FNP + GNP + UNNP \quad (1)$$

where *PD* is the amount of pollutants discharged into the river water, *ULP* and *RLP* are the amounts of pollutants from urban living and rural living, respectively, *AP* is the amount of pollutants from agriculture, *IP* is the amount of industrial pollutants, *TIP* is the amount of tertiary industrial pollutants, *UNP* is the amount of urban non-point source pollutants, *FNP* is the amount of forest land non-point source pollutants, *GNP* and *UNNP* are the grassland and unused land non-point source pollutants. All parameters are in $t \cdot a^{-1}$.

Pollutant discharge intensity is mainly affected by factors such as the investment of environmental protection funding, advanced technology, sewage treatment capacity. The amount of pollutants entering the river is mainly influenced by the discharge intensity and the output coefficient of non-point source pollutants. In this study, data of the discharge intensity of each industry type and urban/rural living were collected from the environmental quality statistical yearbooks and the second national survey of pollution sources.[2] The output coefficients of non-point source pollutants were mainly collected from the published research related to the YRB. The related data are shown in Table S1.

(2) Land use module

Land use can be regarded as an external input in the SD model, as it is derived from the outputs of the CA model. As the direct carrier of human activities, land use directly affects the spatial pattern of the WECC. For example, the loss rate of non-point source pollutants and water resource yield are greatly affected by the type of land use.

2.2.2. Construction of the CA–Markov model

The CA–Markov model includes a CA module and Markov chains. The Markov module is responsible for forecasting the quantitative land use, and the CA module is responsible for simulating the spatial pattern

[2] Considering the base year of the second national survey of pollution sources is 2017, the distribution of water use and pollutants are based on 2017.

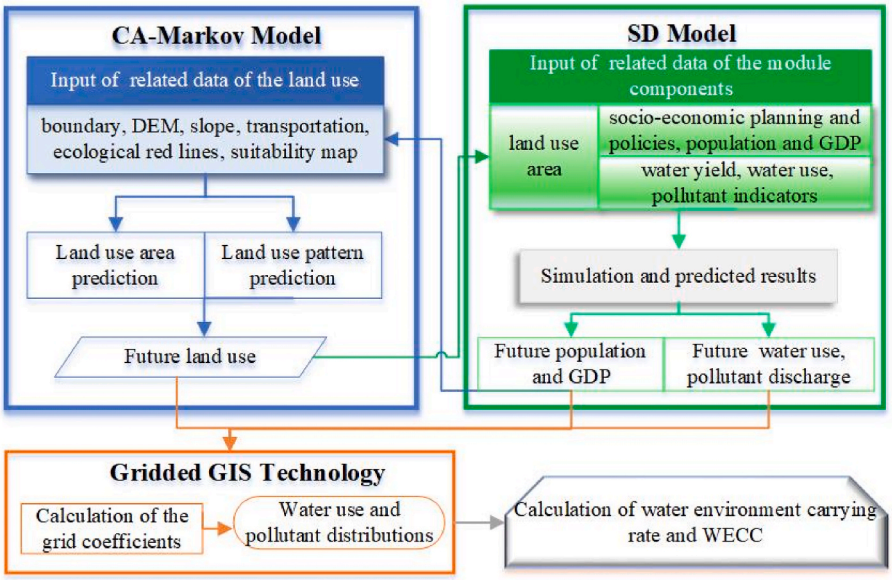


Fig. 2. The framework of the spatial system dynamics model based on the coupled SD, CA, and GIS technology.

changes (Sang et al., 2011). The core part of the CA–Markov model is to produce the transfer matrix and the suitability maps for the land use categories (Fu et al., 2018; Guan et al., 2011).

Owing to urban development differences between Zhangjiakou and Beijing, in the present study we considered these as the boundaries to construct the land use subsystems. First, based on the land use classification of 2000 and 2005, the transfer matrix was generated by the Markov chain. The multi-criteria evaluation module was used to determine which land-use types are suitable for the transformation from one type to another. The factors affecting land use change were divided into two categories, namely driving and constraint factors, and the weight of each influencing factor was calculated using the analytic hierarchy process. Then, the generated transfer matrix and suitability maps were used to predict the land use of 2015, and the predicted and observed land uses were used to test the simulation error. Finally, after error checking and correction, we predicted land use in 2025 and 2035. More details of the CA-Markov simulation process are shown in our previous work (Dai et al., 2020a).

2.2.3. Socio-economy, water use, and water pollution spatial distribution

2.2.3.1. Population and GDP density structure. According to the economic pattern of the Beijing-Tianjin-Hebei region, the urban and rural dual system is obvious, thus land use change can directly reflect the distribution characteristics of the population and the economy (Liu et al., 2016). Considering the research scope and the precision of data, in this study we considered the district as the smallest administrative unit and urban population and rural population numbers in each district as the control variables to calculate and allocate the population coefficient of each grid. The size of the grid was 450 m × 450 m.

(1) Population distribution:

Referring to the existing grid data processing method (Zhou et al., 2017), in this study we assumed that the urban population was distributed in urban land and the rural population was distributed in rural land. The calculation of population distribution based on land use data was based on the following equation:

$$POP_{ij} = pop_{ir} \times coef_{jr} \Big/ \sum_{j=1}^{n} coef_{jr} + pop_{iu} \times coef_{ju} \Big/ \sum_{j=1}^{n} coef_{ju} \quad (2)$$

where POP_{ij} is the population of j grid of district i, pop_{ir} is the rural population of district i, pop_{iu} is the urban population of district i, n is the number of grids of district i, and $coef_{jr}$ and $coef_{ju}$ are the rural population and urban population coefficients, respectively, of grid j. The detailed calculation process of $coef_{jr}$ and $coef_{ju}$ is shown in equations S6 and S7.

(2) Economic distribution:

Similarly, the gridded output value data of different industries were also calculated and allocated based on the land use type. According to the statistical data of the social and economic yearbook of the study area, the grid calculation formula of the output value of different industries was performed based on the following equation:

1) Primary industry:

$$ECON1_{ij} = Econ_{1i} \times coef1_j \Big/ \sum_{j=1}^{n} coef1_j \quad (3)$$

where, $ECON1_{ij}$ is the added value of the primary industry of grid j of district i, 10^4 CNY; $Econ_{1i}$ is the added value of the primary industry of district i, 10^4 CNY; $coef1_j$ is the coefficient of the primary industry added value of grid j. The detailed calculation process of $coef1_j$ is shown in equation S(8). According to the structure and proportion of the primary industry in each district, the primary industry was mainly related to farmland, forestland, and grassland.

2) Secondary industry:

The correlation analysis showed that the second industrial added value had the closest relationship with urban land use (r=0.953, Fig. S2). The secondary industry added value was mainly distributed on the urban land and the grid calculation of the industrial added value was performed according to the following equation:

$$ECON2_{ij} = Econ_{2i} \times coef2_j \Big/ \sum_{j=1}^{n} coef2_j \quad (4)$$

where $ECON2_{ij}$ is the second industrial added value of grid j of district i, 10^4 CNY; $Econ_{2i}$ is the second industrial added value of district i, 10^4 CNY; $coef2_j$ is the coefficient of the second industrial added value of grid j. The detailed calculation process of $coef2_j$ is shown in equation S(9).

3) Tertiary industry:

$$ECON3_{ij} = Econ3_{ir} \times coef3_{jr} \Big/ \sum_{j=1}^{n} coef3_{jr} + Econ3_{iu} \times coef3_{ju} \Big/ \sum_{j=1}^{n} coef3_{ju} \quad (5)$$

where $ECON3_{ij}$ is the added value of the tertiary industry of grid j of district i, 10^4 CNY; $Econ3_{ir}$ is the added value of the rural tertiary industry of district i, 10^4 CNY; $Econ3_{iu}$ is the added value of the urban tertiary industry of district i, 10^4 CNY; $coef3_{jr}$ is the coefficient of the added value of the rural tertiary industry of grid j, $coef3_{ju}$ is the coefficient of the added value of the urban tertiary industry of grid j. The detailed calculation process of $coef3_{jr}$ and $coef3_{ju}$ are shown in equations S10 and S11.

2.2.3.2. Water use structure. According to the gridded population and economy calculation results, the total water use of each grid was determined by the urban and rural population numbers, the scale of different industrial output values, and the intensity of water use.

$$WU_{ij} = \mathrm{wu}_{ir} \times pop_{ir} \times coef_{jr} \Big/ \sum_{j=1}^{n} coef_{jr} + \mathrm{wu}_{iu} \times pop_{iu} \times coef_{ju} \Big/ \sum_{j=1}^{n} coef_{ju} + ECON1_{ij} \times wu_{i1} + ECON2_{ij} \times wu_{i2} + ECON3_{ij} \times wu_{i3} \quad (6)$$

where WU_{ij} is the amount of water use of j grid of district i, m^3; wu_{ir} is the intensity of domestic water use for rural residents of district i, m^3 per capita·a^{-1}; wu_{iu} is the intensity of domestic water use for urban residents of district i, per capita a^{-1}; wu_{i1}, wu_{i2}, and wu_{i3} are the intensity of primary, secondary, and tertiary industry water uses, respectively, of district i, $m^3 \cdot 10^{-4}$ CNY. Other parameters are the same as mentioned in the previous equations. In this equation, we incorporated environmental water use into urban residents' domestic water use, because in this region, most of the environmental water use occurs in the urban residential areas.

2.2.3.3. Total phosphorus density structure. The domestic, primary, industrial, and tertiary industry source pollutants, as well as the non-point source pollutants exported by different land uses, were calculated based on the grid population and economic density and the predicted land use maps. In YRB, the prevalent water pollutant was total phosphorus (TP), the amount of which exceeded the water quality standard in most monitoring sites. Therefore, we mainly focused on TP pollutants.

1) Inflow of TP from domestic pollutants:

$$DTP_{ij} = tp_{ir} \times coef_{jr} \Big/ \sum_{j=1}^{n} coef_{jr} + tp_{iu} \times coef_{ju} \Big/ \sum_{j=1}^{n} coef_{ju} \quad (7)$$

where DTP_{ij} is the inflow of TP from living sources of j grid of district i, t; tp_{ir} is the inflow from rural residents of district i, t; tp_{iu} is the inflow of urban residents of district i, t; other parameters are the same as mentioned above.

2) Inflow of TP from industrial pollutants:

Similar to the grid calculation process of the second industrial added value, according to the inflow amount of TP of the industrial enterprises, the gridded industrial TP inflow was calculated as follows:

$$ITP_{ij} = itp_i \times ucoef_j \Big/ \sum_{j=1}^{n} ucoef_j \quad (8)$$

where ITP_{ij} is the inflow amount of industrial pollutants of grid j of district i, t; itp_i is the inflow amount of industrial pollutants of district i, t; $ucoef_j$ is the urban land use coefficient of grid j. In this study, the industrial pollution load only came from urban land. The detailed calculation process of $ucoef_j$ is shown in equation S(12).

3) Inflow of TP from tertiary industry pollutants:

$$TP3_{ij} = tp3_{ir} \times coef3_{jr} \Big/ \sum_{j=1}^{n} coe3f_{jr} + tp3_{iu} \times coef3_{ju} \Big/ \sum_{j=1}^{n} coe3f_{ju} \quad (9)$$

where $TP3_{ij}$ is the inflow amount of TP from the added value of the tertiary industry of grid j of district i, t; $tp3_{ir}$ and $tp3_{iu}$ are the inflow amounts of TP from the added value of the tertiary industry in rural land and urban land, respectively, of district i, t. The other parameters are the same as above.

4) Inflow of TP from planting pollutants:

$$PTP_{ij} = ptp_i \times facoef_j \Big/ \sum_{j=1}^{n} facoef_j \quad (10)$$

where PTP_{ij} is the inflow amount of TP from the planting of grid j of district i, t; ptp_i is the inflow amount of TP of district i, t; $facoef_j$ is the farmland coefficient of grid j. In this study, the pollution load of planting only came from farmland. The detailed calculation process of $facoef_j$ is shown in equation S(13).

5) Inflow of TP from livestock and poultry breeding pollutants:

According to the scale of livestock and poultry breeding enterprises and the TP discharge amount, the gridded inflow density of the TP of livestock and poultry breeding was calculated as follows:

$$BTP_{ij} = btp_i \times rcoef_j \Big/ \sum_{j=1}^{n} rcoef_j \quad (11)$$

where BTP_{ij} is the amount of TP from livestock and poultry breeding of grid j of district i, t; btp_i is the amount of TP from livestock and poultry breeding of district i, t; $rcoef_j$ is the rural land use coefficient of grid j. As in this region, the current large-scale livestock and poultry breeding manure is almost all dry-treated and converted into organic fertilizer, the TP discharged from breeding only comes from the scattered feeding of livestock and poultry on rural land. The detailed calculation process of $rcoef_j$ is shown in equation S(14).

6) Inflow of TP from other non-point sources:

$$OTP_{ij} = ftp_i \times focoef_j \Big/ \sum_{j=1}^{n} fcoef_j + utp_i \times ucoef_j \Big/ \sum_{j=1}^{n} ucoef_j + gtp_i \times gcoef_j \Big/ \sum_{j=1}^{n} gcoef_j + +ntp_i \times uncoef_j \Big/ \sum_{j=1}^{n} uncoef_j \quad (12)$$

where OTP_{ij} is the inflow amount of non-point source TP from other land use types of grid j of district i, t; ftp_i is the inflow amount of TP from forest land of district i, t; utp_i is the inflow amount of TP from urban land of district i, t; gtp_i is the inflow amount of TP from grassland of district i, t; $untp_i$ is the inflow amount of TP from unused land of district i, t; $focoef_j$ is the forest land coefficient of grid j, $ucoef_j$ is the urban land use coefficient of grid j; $gcoef_j$ is the grassland coefficient of grid j; $uncoef_j$ is the unused land coefficient of grid j. The detailed calculation processes of $focoef_j$, $gcoef_j$, and $uncoef_j$ are shown in equations S15-S17.

2.3. Water environment capacity

To better reveal the natural endowment of the Yondging River water environment, the water environment capacity (WEC) at the control unit scale was calculated, as the control unit can express the spatial correlation between the water environment pressures and the WEC (Zhang et al., 2021; Zhou et al., 2017). Owing to the small river widths, the one-dimensional steady-state water quality model QUAL2K was used to calculate the WEC of each control unit (Zhang et al., 2012). The QUAL2K model regards the river as a one-dimensional channel with a steady flow and considers the effects of point source and non-point source pollution loads (Zhang et al., 2012). It consists of four parts, the division of river segments, the calculation of river pollution loads, the calibration of hydraulics, and water quality parameters. Finally, the simulated river water quality was used to calculate the WEC. The river WEC was calculated as follows:

$$WEC = 86.4(C_N - C_0)Q + KC_N\left(\frac{L}{U}\right)Q \tag{13}$$

where WEC is the water environment capacity of the control unit, kg·d^{-1}; C_N is the water quality target, mg·L^{-1}; C_0 is the river water pollutant concentration simulated by the QUAL2K model, mg·L^{-1}; Q is the river flow m^3·s^{-1}, which corresponding to 75% hydrological frequencies of the driest month in recent 10 years (Sun et al., 2021); K is the pollutant degradation rate; the organic phosphorus hydrolysis is 0.2–0.4 d^{-1}; and the organic and inorganic phosphorus settling velocities are 1.0 m d^{-1} and 0.5 m d^{-1}, respectively (Li et al., 2010); L is the river length, km; U is the average flow velocity, m·s^{-1}.

2.4. Water environment carrying rate and WECC

2.4.1. Calculation of water environment carrying rate

According to the concept of WECC, the quotient of the water environment pressure, i.e., the amount of water use and pollutant discharge, and the supporting factors, i.e., the available water resources and the available WEC were used as the carrying rate of the water environment system. Based on the Cannikin law, the water environment carrying rate (WECR) is the minimum value of the WRCR and WPCR. It was calculated using the following equation:

$$WECR = min(WRCR,\ WPCR) \tag{14}$$

where $WECR$ is the water environment carrying rate. $WRCR$ is calculated by the quotient of the total amount of available water resources and the total water use in each control unit. According to the predicted water yield based on the land use type and the proportion of each land use category, the water resources of each control unit are allocated. $WPCR$ is calculated by the quotient of the available WEC and the total amount of pollutants discharged into each control unit. The minimum value is chosen as the $WECR$ to express the water crisis in YRB. When $WECR > 1$, the water environment system is sustainable for socio-economic development; when $WECR < 1$, the water environment is in overloaded status; when the index of $WECR = 1$, the socio-economic development, and the natural water environment system are in balance and there is no water crisis.

2.4.2. WECC calculation

The essence of the WECC calculation is to determine the maximum population and economic scale that a healthy water environment system can carry. During the calculation of WECC, a key problem that must be tackled is under what state or what conditions can the water environment system carry socio-economic development. Ignoring other carrying capacities of the water environment system, i.e., shipping, aesthetics, entertainment, we only selected two pressure and two supporting factors to calculate the WECC.

(1) WECC calculation based on water use and available water resources

Based on the calculation process of the water use, when the total amount of water use is equal to the available water resources (AWR), the equation for the population that the water resources system can carry is as follows:

$$POP_W = \frac{W_a - W_{en} - (GDP_1 \times Q_{w1} + GDP_2 \times Q_{w2} + GDP_3 \times Q_{w3})}{UR \times Q_{wu} + (1 - UR) \times Q_{wr}} \tag{15}$$

where POP_W is the total population the water resource system can carry, 10^4; w_a and W_{en} are the AWR and the environment water use, respectively, 10^4 m^3; UR is the urbanization rate; Q_{wu} is the domestic water quota of urban residents, m^3 per capita·a^{-1}; Q_{wr} is the domestic water quota of rural residents, m^3 per capita·a^{-1}; Q_{w1}, Q_{w2} and Q_{w3} represent the water use intensity per unit added values of the primary, secondary, and tertiary industries, respectively, m^3·10^{-4} CNY. GDP_1, GDP_2, and GDP_3 are the added values of the primary, secondary and tertiary industries, respectively, 10^4 CNY.

Similarly, when the total water use is equal to the AWR, the equation of GDP that the water resources system can carry is as follows:

$$GDP_W = \frac{W_a - w_{en} - [POP_W \times UR \times Q_{wu} + POP_W \times (1 - UR) \times Q_{wr}]}{a \times Q_{w1} + b \times Q_{w2} + c \times Q_{w3}} \tag{16}$$

where GDP_W is the total GDP the water resource system can carry, 10^4 CNY; a, b, and c are the ratios of the GDP_1, GDP_2 and GDP_3 industries, respectively, to the total GDP; other parameters are the same as above.

(2) WECC calculation based on the amount of pollutants discharged and the available WEC

According to the calculation process of pollutants discharged into the river, when the pollutant discharge is equal to the available WEC, the calculation equation of the population that the water environment system can carry is as follows:

$$POP_e = \frac{AWEC - (GDP_1 \times P_{d1} + GDP_2 \times P_{d2} + GDP_3 \times P_{d3}) - UNP - FNP - ONP}{UR \times P_{du} + (1 - UR) \times P_{dr}} \tag{17}$$

Table 1
The error test result of the SD model.

Sub-regions	Year	People (10^4)			GDP (10^8 CNY)			Water use (10^8 m^3)			Wastewater discharge (10^8 m^3)		
		Simulated	Observed	Error (%)	Simulated	Observed	Error (%)	Simulated	Observed	Error (%)	Simulated	Observed	Error (%)
Beijing	2014	62.20	61.9	**0.48**	233.63	236.35	**1.15**	9988	10630	**6.04**	4013	3648	**10.01**
	2015	62.20	65.6	**5.18**	251.43	258.14	**2.60**	9722	10230	**4.96**	4010	3736	**7.33**
	2016	63.80	66.42	**3.94**	280.52	292.06	**3.95**	9803	10500	**6.63**	4146	4022	**3.09**
	2017	66.20	68.6	**3.50**	310.56	319.60	**2.83**	9903	10580	**6.40**	4338	4095	**5.93**
Zhangjiakou	2014	351.42	353.89	**0.70**	975.20	1049.92	**7.12**	74721	72974	**2.39**	10511	11659	**9.84**
	2015	351.76	322.14	**9.19**	930.59	990.27	**6.03**	74722	73475	**1.70**	10655	12408	**14.13**
	2016	352.20	329.73	**6.81**	1023.66	1068.59	**4.20**	72086	71483	**0.84**	10513	11369	**7.53**
	2017	352.66	344.06	**2.50**	1091.01	1207.98	**9.68**	72430	70945	**2.09**	10696	12297	**13.02**

where POP_e is the total population the available water environment capacity can carry, 10^4; $AWEC$ is the available WEC, t; P_{du} is the inflow intensity of pollutants from urban residents; P_{dr} is the inflow intensity of pollutants from rural residents; P_{d1}, P_{d2}, and P_{d3} are the inflow intensities of pollutants of the per unit added value of the primary, secondary, and tertiary industries, respectively, $t{\cdot}10^{-4}$ CNY; UNP is the inflow of urban non-point source pollutants, t; FNP is the inflow of forest land non-point source pollutants, t; ONP is the inflow of other land use non-point source pollutants (e.g., grassland and unused land); other parameters are the same as above.

Similarly, when the total amount of the inflow of pollutants is equal to the AWEC, the equation of GDP that the available water environment capacity can carry is as follows:

$$GDP_e = \frac{AWEC - UNP - FNP - ONP - [POP \times UR \times P_{du} + POP \times (1 - UR) \times P_{dr}]}{a \times P_{d1} + b \times P_{d2} + c \times P_{d3}} \quad (18)$$

where GDP_e is the total GDP the available water environment capacity can carry, 10^4 CNY; other parameters are the same as above. Finally, the Cannikin law is used to calculate the results of the WECC, that is, the minimum population and GDP results are the final WECC.

3. Results and discussion

3.1. Model calibration and validation

3.1.1. SD model

In the SD model, the observed data from 2006 to 2013 obtained from Zhangjiakou and Beijing were used to calibrate and validate the parameters. The simulation accuracy was verified using data from 2014 to 2017 (Table 1). The relative errors of the simulation results for the population and GDP[3] and the other major variables from 2014 to 2017 were below 15% (Table 1), indicating that the SD model performed well. Therefore, the SD model can be used to simulate the human and water environment systems. The predicted results of the future water use and pollution discharge of the two sub-regions are shown in Fig. S3 and Fig. S4.

3.1.2. CA model

Based on the latest research, three land use change components ("quantity," "exchange," and "shift") (Pontius and Santacruz, 2014; Quan et al., 2018) were chosen to verify the reliability of the land use prediction results (Fig. 3). Compared with the quantity and shift components, the prediction error of the exchange component was larger, ranging from 0.02% (unused land) to 0.91% (farmland) of the domain (Fig. 3a). The intensity analysis showed that the intensities were active for urban and rural land gains, and unused land loss, while dormant for the forestland (Fig. 3b). Compared with the observed land use, the simulated unused land had the largest gain (13.84%). This was followed by the farmland (3.63%), while rural land (10.04%) had the largest loss intensity, followed by urban land (8.40%), thereby indicating that urban land and unused land had higher simulation errors in the entire spatial extent. In general, the land use change components and the changing intensity of the seven land use types were all lower than 1.8% and 14.8%, respectively. This indicated that the CA-Markov model performed well in the land use prediction of the YRB. Subsequently, the predicted land use data were used in the SD model to simulate the water resources yield and non-point source pollutant export.

As is shown in Fig. 4, most of the new urban land was expanded around the existing urban land. From 2015 to 2035, the urban land in Zhangjiakou will increase from 558.74 km^2 to 673.09 km^2. This is mainly due to the continuous increase in the urbanization rate of Zhangjiakou. Meanwhile, due to the implementation of the water conservation area protection and control of the "city disease" policy, urban land expansion in Beijing's mountainous areas has been slowed down, with an increased area of 18.29 km^2. Based on the Yongding River restoration plans, in the process of land use simulation, we assumed that the shrinking water area would gradually recover to the level of 2000. Thus, the water area of Zhangjiakou will increase from 285.12 km^2 to 296.23 km^2 and the water area of Beijing's mountainous area will increase from 21.81 km^2 to 31.54 km^2 in 2035 (Fig. 4). The increase of the water area will promote the yield of the local water resources. However, the increase of the urban land may generate more urban non-point sources, which will adversely affect the river water quality.

3.2. Temporal and spatial variation of water use and TP pollution

The spatial pattern of water use and TP pollution in 2017, 2025, and 2035 directly indicate that water use intensity increases and TP discharge intensity decrease as time progresses (Fig. 5), especially in the urban regions with high population and GDP (Fig. S5). In 2017, the total amount of water use was 8.20×10^8 m^3 (1.00×10^8 m^3 in Beijing's mountainous area, and 7.20×10^8 m^3 in Zhangjiakou; Fig. S3). Agricultural water use was the greatest, and accounted for 66.88% of the total water use, followed by domestic water use (17.75%). This is mainly because of the limited precipitation, which forces agricultural production to be heavily dependent on irrigation. Simultaneously, the total amount of TP discharge was 161.01 tons (138.30 tons in Zhangjiakou

[3] The average exchange rate in 2021 for one CNY is 0.154 USD.

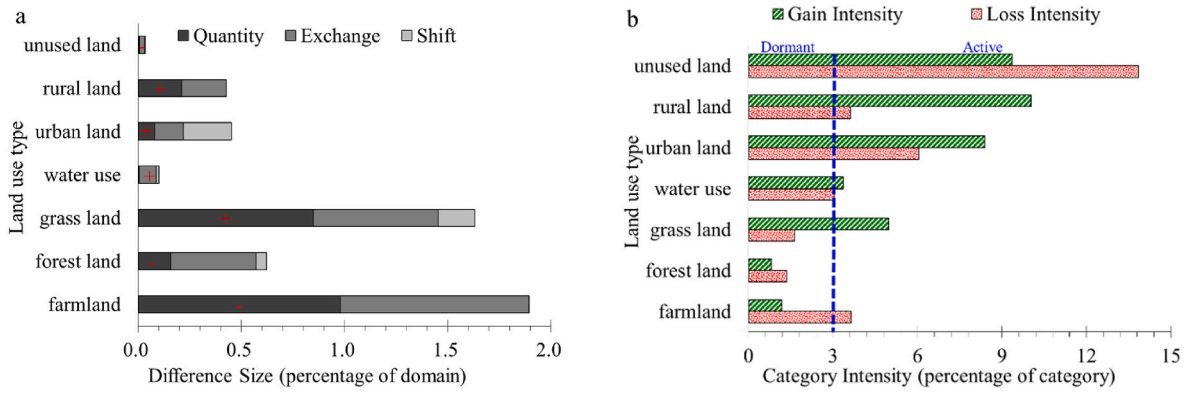


Fig. 3. Comparison of the land use type changing (a) and the changing intensity (b) between the observed and the simulated land use of 2015 ('+' indicates a net increase and '−' indicates net decrease; the dashed vertical line is the uniform intensity of the change for the entire spatial extent).

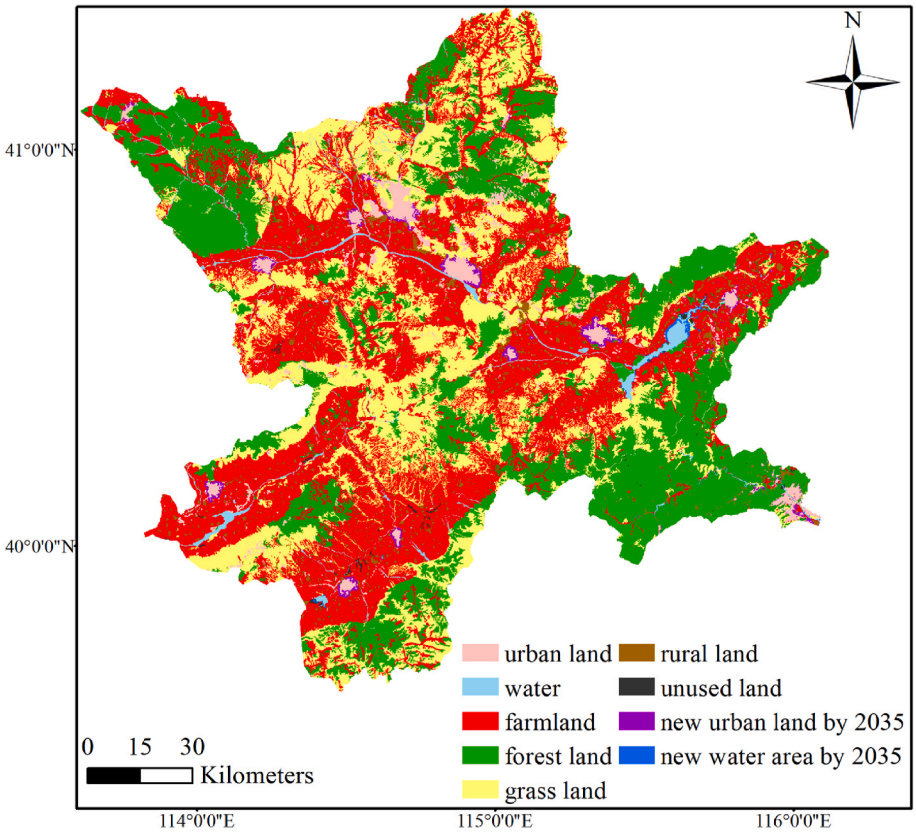


Fig. 4. Current land use map of 2015 and the predicted urban land and water area of 2035 in the study area.

and 22.71 tons in Beijing's mountainous area; Fig. S4). The largest source of TP discharge was the urban living source (33.97%), followed by the rural living source (28.94%) (Fig. S4).

Based on the water use map (Fig. 5a), it was obvious that the spatial pattern of water use intensity was contrary to the water use amount; in other words, the agriculture water use intensity was lower than urban living and secondary industry water use intensities. The water use per area of farmland (6.67×10^4 m^3 km^{-2}) was much lower than that of urban land (36.61×10^4 m^3 km^{-2}). The urban area is usually the water use hotspot area, due to the densely populated and industry cluster areas, e.g., control units 14 and 15 in Zhangjiakou and control units 4 and 13 in Beijing (Fig. 4). However, the TP discharge intensity was similar to both areas (Fig. 5d). Urban land had the largest discharge intensity (122.40 kg km^{-2}), followed by rural land (114.04 kg km^{-2}).

In 2025, the total water use amount will increase to 8.50×10^8 m^3 along with the increased population and GDP (Fig. S5). Agriculture water use will still be the highest (5.28×10^8 m^3), while it will decrease by 0.21×10^8 m^3 compared with that in 2017 (Fig. S3). The declined agricultural use will quickly be offset by an increase in domestic, industrial, and tertiary industry water use. In 2035, the total water use amount will continue to increase to 8.83×10^8 m^3 (Fig. S3). Compared with 2017, the water use intensity of the entire urban area will increase by 8.56×10^4 m^3 km^{-2} (Fig. 5c). However, urban living TP discharge initially shows an increasing trend, followed by a decreasing trend (Fig. S4), thereby stipulating that the regional socio-economic development follows the Environmental Kuznets Curve (Grossman and Krueger, 1995). The main reason for the decline is the improvement of the capacity of the wastewater treatment plants (Cheng et al., 2018; Ma et al., 2020). Socio-economic development and technological progress are the main driving factors that affect the TP discharge amount. In

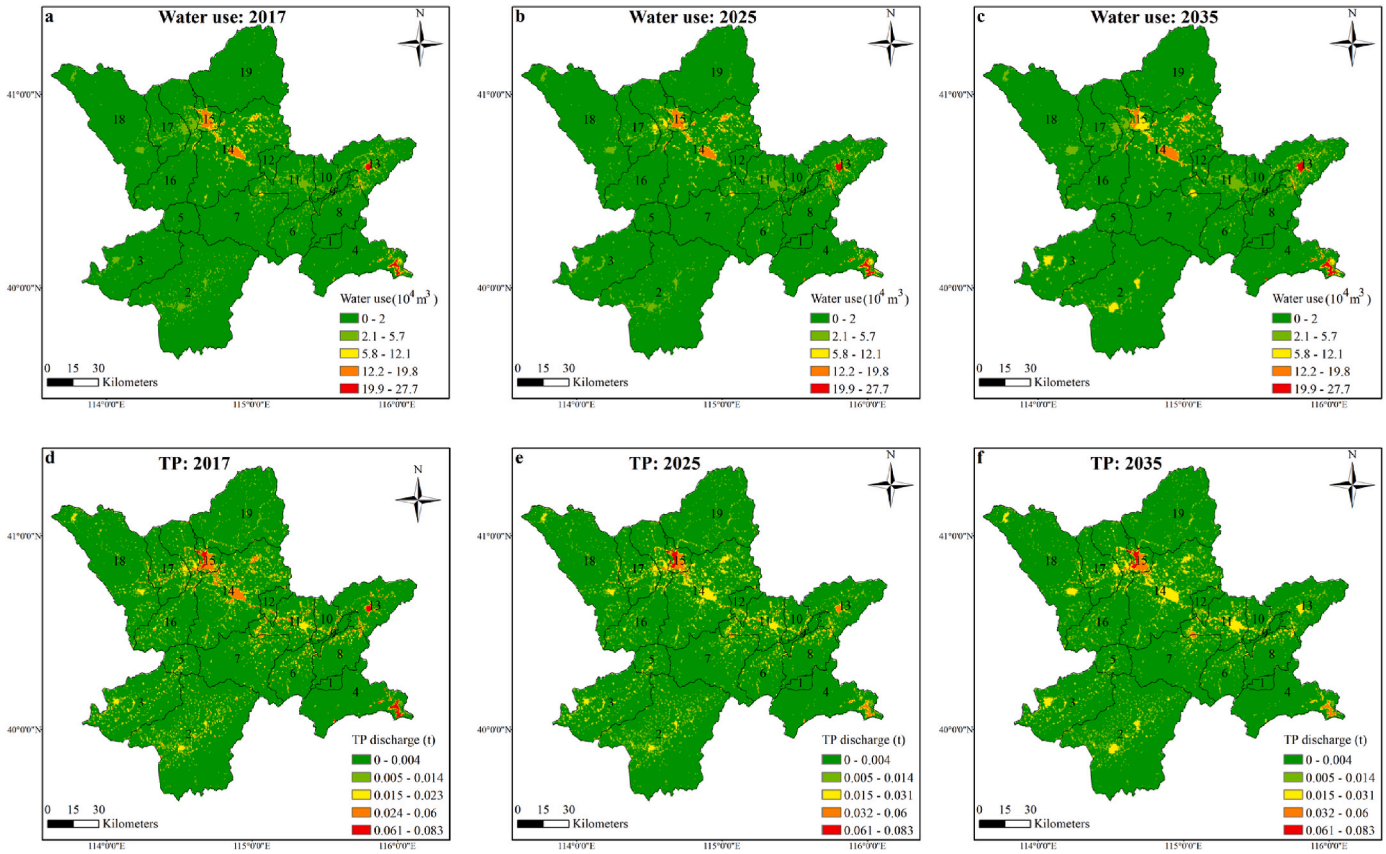


Fig. 5. Spatial distribution of water use (a, b, c) and TP discharge (d, e, f) in the study area.

Table 2
AWEC of TP, AWR, water use, TP discharge, and WECR of each control unit.

Sub-regions	Control units	AWEC of TP	AWR (10^4 m^3)			Water use (10^4 m^3)			TP discharge (t)			WECR		
			2017	2025	2035	2017	2025	2035	2017	2025	2035	2017	2025	2035
Beijing	1	8.81	503.59	497.90	542.49	48.13	36.65	36.10	0.07	0.03	0.03	10.46	13.58	15.03
	4	21.27	6333.84	6147.82	6324.33	4416.89	5054.82	5832.47	12.18	8.11	7.43	1.43	1.22	1.08
	13	2.23	4738.08	4798.98	5602.88	5606.02	5738.58	6512.07	10.46	7.45	6.91	0.21	0.30	0.32
Zhangjiakou	2	13.05	6842.53	6674.66	7418.53	12440.72	12440.53	12511.66	17.98	15.42	14.87	0.55	0.54	0.59
	3	13.56	3142.91	3065.80	3407.47	6533.21	6537.61	6950.20	9.34	8.81	8.62	0.48	0.47	0.49
	5	7.17	728.01	710.15	789.29	1122.05	1087.95	1057.80	1.82	1.59	1.46	0.65	0.65	0.75
	6	2.77	1461.47	1425.61	1584.49	2687.42	2678.31	2496.63	3.92	3.72	3.25	0.54	0.53	0.63
	7	21.34	3026.27	2952.02	3281.02	3653.49	3670.19	3493.78	5.39	4.83	4.61	0.83	0.80	0.94
	8	9.30	1454.89	1419.19	1577.36	1632.20	1621.36	1573.11	4.34	3.92	3.80	0.89	0.88	1.00
	9	–	–	–	–	–	–	–	–	–	–	–	–	–
	10	3.00	670.98	654.52	727.47	1202.46	1193.87	1164.43	3.52	3.19	2.96	0.56	0.55	0.62
	11	4.53	1615.24	1575.61	1751.21	3774.81	3927.95	4426.27	12.40	12.35	13.43	0.37	0.37	0.34
	12	2.06	480.73	468.94	521.20	838.35	843.98	927.39	2.10	1.99	1.87	0.57	0.56	0.56
	14	11.42	4774.32	4657.19	5176.22	14776.94	16114.81	17664.89	29.93	27.77	27.15	0.32	0.29	0.29
	15	6.96	472.23	460.64	511.98	4334.82	4849.54	4887.56	15.97	18.67	18.60	0.11	0.09	0.10
	16	2.23	2969.13	2896.29	3219.07	5632.52	5670.08	5426.43	10.52	10.07	9.62	0.21	0.22	0.23
	17	4.63	1473.07	1436.93	1597.07	3146.14	3203.34	2942.91	7.34	6.92	6.81	0.47	0.45	0.54
	18	4.94	4823.10	4704.77	5229.11	5998.35	6133.67	6157.45	8.73	8.42	8.50	0.57	0.59	0.58
	19	3.68	4199.20	4096.17	4552.68	4185.67	4189.12	4275.48	5.00	4.49	4.36	0.74	0.82	0.84
	Total	142.96	49709.58	48643.20	53813.87	82030.20	84992.37	88336.62	161.01	147.75	144.28	–	–	–

2025, the TP amount will decrease to 147.75 tons and will continue decreasing to 144.28 tons in 2035 (Fig. S4). Notably, tertiary industry and urban non-point TP discharge increase owing to the increased urban extension and GDP (Fig. S5).

3.3. Dynamic evaluation of the WECR under the control unit scale

Considering that WECC research focuses on the specific region or river/lake basin instead of the grid (Hu et al., 2021; Zhou et al., 2017), as

Table 3
Comparison between the calculated WECC and the population, GDP of the study area in YRB.

Year	Sub-regions	Calculated WECC				Actual and predicted	
		AWR		AWEC of TP			
		People (10^4)	GDP (10^8 CNY)	People (10^4)	GDP (10^8 CNY)	People (10^4)	GDP (10^8 CNY)
2017	Zhangjiakou	344.06	441.53	344.06	489.12	344.06	1091.01
	Beijing	86.24	394.69	70.78	378.13	68.60	310.56
2025	Zhangjiakou	356.34	593.16	356.34	689.90	356.34	1826.87
	Beijing	75.45	536.86	71.80	804.37	69.34	476.97
2035	Zhangjiakou	361.00	1031.06	361.00	1610.74	361.00	3487.60
	Beijing	71.45	821.85	91.70	2559.66	71.45	808.79

mentioned above, we introduced a control unit to the calculations of WRCR, TP carrying rate (TPCR), and WECR[4] (Table 2, Fig. S6). The total water use in 2017, 2025, and 2035 all exceed the AWR (Table 2). In 2017, the total discharge amount of TP exceeded its AWEC, while in 2025 and 2035, TP discharge is less than the AWEC of TP (Table 2).

The spatial pattern of the WRCR (Fig. S6a) showed that except for control units 1 and 4 in Beijing's mountainous area and control unit 19 in Zhangjiakou, the current water resources were seriously overloaded, especially control units 14 and 15, thereby indicating severe water shortage in Zhangjiakou. In 2025, the water resources of all the control units will still be overloaded in Zhangjiakou (Fig. S6b). It is worth noting that control unit 15 had the lowest WRCR (0.09) (Fig. S6b). This is mainly because the region has the highest urbanization rate and the highest water use intensity (18.3×10^4 m^3 km^{-2}). By 2035, the overloading conditions of water resources will be improved (Fig. S6c), which is mainly related to the water-saving measures and the increased reclaimed water supply under the current developing level. In terms of TP discharge, TPCR was much higher than the WRCR (Fig. S6). In 2017 and 2025, only seven control units are surplus with the AWEC of TP, and this will increase to eight in 2035 (Figs. S6d–f).

The WECR showed that except for control units 1, 4, and 8, from 2017 to 2035 the water environment system would continue to overload, with water shortage being the dominant cause (Table 2). The overloaded control units 13, 14, and 15 show that densely populated and economically developed areas are often heavily overloaded (Table 2). Therefore, in these areas, the water use intensity should decrease and the TP treatment efficiency should improve. The agricultural irrigation areas and rural villages' gathering areas, such as control units 2, 3, 16, 17, and 18, need to improve their utilization of irrigation water and fertilizer and decrease the living, planting, and livestock and poultry breeding TP discharge.

The WECR calculation results show that the mismatching between AWR, and AWEC, induced by natural geographic conditions, and the water use and TP discharge, induced by the socio-economic development are quite obvious in YRB, although the hydraulic engineering construction, advanced water-saving technology, and sewage treatment plans could narrow the gap between available water resources, water use, and TP discharge. From the perspective of the spatial distribution of WRCR and TPCR (Fig. S6), during the specific period (2017, 2025 and 2035), nine control units (2, 6, 11, 13 and 14 to 18) are overloaded with both the AWR and AWEC and seven control units are overloaded with either AWR or AWEC, while control units 1 and 4 show a surplus of both AWR and WEC. These results further evidence the spatial heterogeneity of the WECC across the basin. In this case, when making decisions for socio-economic development, it is necessary to fully consider the spatial effects, and the AWR and AWEC should be set as limiting factors.

Many rivers in North China are drying up, mainly due to the increasing water demand (Chen and Sun, 2015; Quan et al., 2018; Zhao et al., 2010, 2015). In the past 30 years, the surface runoff of Yongding River has decreased while the average annual precipitation did not show a significant downward trend (Dai et al., 2020b; Jiang et al., 2014), indicating that precipitation has a small impact on the decline of natural river runoff. Existing studies have shown that the over-extraction of groundwater, agricultural water irrigation, reservoir interception, and impoundment are the main factors influencing the drying up of rivers in the YRB (Hou et al., 2019; Jiang et al., 2014). Therefore, the overuse of water resources is the main cause of the low WRCR.

Owing to the dry climatic conditions and year-round droughts in YRB, farmland has been modified as a water-holding system to fully absorb the rainwater instead of allowing runoff to flow downstream freely. Studies have shown that except for extreme rainstorms, it is almost difficult to form a continuous surface runoff in the semi-arid regions of North China (Ongley et al., 2010; Pernet-Coudrier et al., 2012; Yu et al., 2008). In the rural areas of the Hai River Basin, only rainfall events above 80 mm can form runoff into rivers and/or ditches, but such rainstorm events occur almost once every 5 years (Liu et al., 2006; Pernet-Coudrier et al., 2012). This indicates that farmland runoff is generally rare in RYB. In contrast to the development and utilization of rivers in Europe and the United States, most of the rivers in China have been constructed with dams and reservoirs. Therefore, these artificial facilities have interfered greatly with the natural hydrological processes. During the water storage period, a large number of point source pollutants are accumulated. These accumulated pollutants will only be released into downstream water during high flow events, such as rainstorms. In downstream rivers, the release of these cumulative pollutants appears to be a non-point source "runoff scouring effect" (Yu et al., 2008); in fact, they are point sources. Guan et al. (2014) pointed out that cumulative water pollution is the key driving factor for the water scarcity across China caused by water pollution. This is also consistent with the TP load structure in our study area, that is, urban living (point source) is the major source of TP (Fig. S4). Furthermore, at present, the rivers in YRB mainly rely on the effluent from the wastewater treatment plants and the discharge of the upstream reservoir for replenishment (Cheng et al., 2018) resulting in insufficient dilution and the self-purification capacity of TP in the river water, such as control units 13, 14 and 15.

3.4. Threshold of the WECC measured by population and GDP

According to the WECR results, when the water environment system is overloaded, priority should be given to the carrying capacity of the population, followed by economic development. Based on the equations of WECC in Section 2.4.2, the population and economic scales are calculated by finding the critical point where the AWR and AWEC of TP are equal to the total water use and TP discharge of the two sub-regions, that is, the WECC measured by population and GDP. Considering that Zhangjiakou's water resources are significantly overloaded, in this case,

[4] Considering the uncertainty of the application of water quality models to calculate the WEC and the fluctuation characteristics of the real water quality, a certain amount of WEC needs to be reserved as a margin of safety (MOS). In this study, 5% of the WEC of TP is taken as the safety margin to meet the assimilate capacity of the water environment system. Therefore, 95% of the WEC of TP; that is the available WEC of TP, is involved in the TPCR calculation in Table 2 and Fig. S6.

the actual population should be used as the population threshold of the WECC, thus priority should be given to guaranteeing domestic water supply and pollution assimilation; then, the rest of the AWR and the AWEC of TP can be used to support economic development.

The results of this study showed that in 2017, the WECC in Zhangjiakou was 3.44×10^6 people and 4.42×10^{10} CNY (Table 3). The current economic scale of the Zhangjiakou area is 1.47-fold higher than the water resources carrying capacity and 1.23-fold higher than the AWEC of TP. The industrial structure in Zhangjiakou is unreasonable, and economic development is at the cost of the over-exploitation of water resources followed by the AWEC. From the perspective of scale, although the WECC of Beijing's mountainous area is higher than the observed population and economy, it is urgent to control TP pollution in control unit 13. In 2025, the overloaded economic scale of the entire area will be 1.04-fold higher than the WECC (Table 3). In 2035, the WECC of Zhangjiakou and Beijing mountainous areas will be 3.61×10^6 people and 1.03×10^{11} CNY and 7.14×10^5 people and 8.22×10^{10} CNY, respectively (Table 3). The economic scale of the entire area will be 1.59-fold higher than the WECC. The calculation of the WECC threshold further shows that the main factor affecting the WECC of the two sub-regions is water shortage, followed by the insufficient AWEC of TP. From the perspective of the WECC scale, the WECC of Beijing's mountainous area is much better than that of Zhangjiakou, although the phosphorus pollution of control unit 13 still needs to be improved. The WECC results corroborated the view of Arrow et al. (1995) that the carrying capacity in nature is not fixed, static, and/or characterized by simple relationships. Overall, the overload situation of water pollution is less than that of water resources, and water shortage is mainly responsible for the overload of WECC in YRB. This is especially true for Zhangjiakou, where water shortage is the main constraint of economic and social development instead of water quality. The impact of water resources on WECC is stronger than that of the AWEC. Currently, to some extent, improving water quantity is more difficult than improving water quality in YRB.

3.5. Measures for eliminating the overloaded conditions

The water problem in YRB is a combination of water shortage and water quality degradation. Improving water resources will promote the improvement of AWEC and improving water quality will, in turn, promote the supply of water resources. In this case, water transfer seems to be the best way to increase the WECC. However, studies have shown that water diversion projects are not enough to solve the water shortage problem in the Hai River Basin (Liu and Yang, 2012; Zhao et al., 2010). In the long run, water shortage still needs to be tackled by irrigation water-saving technologies, planting structure adjustments, increasing the reclaimed water supply, water transfer, and virtual water importing in YRB. From the perspective of water quality, for the overloaded urban areas, such as control units 14 and 15 in Zhangjiakou, the current priority is to improve the wastewater treatment capacity and further reduce point source pollution. At the same time, the overloaded control units in rural areas require improvements in the treatment capacity of rural domestic sewage, and the chemical fertilizer utilization rate to control the non-point source pollution.

4. Conclusions

In this study, an improved spatial system dynamics model integrated with the SD model, CA model, and gridded GIS technology is put forward to simulate the relationship of coupled human and water environment systems. We validated this integrated model and applied it to evaluate the spatiotemporal dynamic changes of the WECC of the YRB in the 2017–2035 period. The main results are summarized below.

Inadequate water resources are mainly responsible for the overloaded WECC in YRB, followed by water pollution. This overload is unevenly distributed within the river basin. From 2017 to 2035, Zhangjiakou often suffers an overload of WECC throughout the year owing to quantity-related water shortages, especially in urban areas. However, in Beijing's mountainous area, the WECC of control units 1 and 4 are in good status owing to sufficient water quantities.

Based on the reasons for the overload, effective and targeted water management measures are provided under the control unit scale. In industrial clusters and densely populated areas with overloaded WECC status, such as control units 13, 14 and 15, cleaner production, water-saving, and wastewater treatment technologies need to be vigorously promoted. In agriculture-dominated areas, i.e., control units 2, 3, 6, 10 and 11, water-saving irrigation and advanced rural wastewater treatment technologies need to be improved.

The SSD model revealed the feedback interactions between human activities and water environment systems and realized the simulation of the spatial heterogeneity of WECC. Given the limitations of previous WECC assessment results, which focused on the dimensionless index assessment, this study proposes a WECC calculation method that is based on population and the GDP and verifies that the threshold of WECC is not static. Our model improves the shortcomings of static and global assessment methods of WECC and the low reliability of the WECC assessment results.

Inevitably, some deficiencies need to be improved in future research. First, the WECC predicted results are only based on current policies and development plans. In the real world, the socio-economic development-induced water use, land use change, water flow, and pollution discharge are complex, changeable, and not easy to calculate and predict accurately in this integrated model. Additionally, the lack of statistical data at the township level could reduce the accuracy of the water use and pollution discharge spatial pattern results. Although the WECC analysis in this study was only based on a baseline scenario, this model could provide the basis for many improvement measures of WECC based on a range of reasonable regulating scenarios resulting from future research.

CRediT authorship contribution statement

Dan Dai: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft, Writing, Visualization, Writing – review & editing. **Mingdong Sun:** Writing – review & editing. **Xubo Lv:** Data curation. **Juxiang Hu:** Writing – review & editing. **Hua Zhang:** Writing – review & editing. **Xiangqin Xu:** Writing – review & editing. **Kun Lei:** Methodology, Supervision, Resources, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jclepro.2022.131137.

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